

Refinery Charge Scheduling I — Freeing the Dewaxing Unit

Objective and Prerequisites

This model decides **what the MEK dewaxing unit charges on each day of a 42-day horizon**, and how much, so that the refinery serves as much demand as it can while keeping every tank inside its physical limits.

It is the second step of a staged build. The first step (v0) took the planner's choice of *what* each unit charges as given and optimised only *how much*, plus where the resulting overflow went. This step frees that choice on one unit — the one whose transition structure is cleanest — and is therefore the first model in the series to contain any binary variables at all.

You should be comfortable with linear programming and with mixed-integer formulations of scheduling problems. Familiarity with the unit commitment problem will help, because the transition structure here is the same one.

What You Will Learn

- How to model a **restricted changeover structure** without an $n \times n$ transition matrix, by creating variables only for the arcs that exist.
- How to carry a machine's **setup state through idle days**, so that a unit cannot dodge a cleanout by shutting down for a day and restarting on something else.
- How to allocate a day's **time** rather than the day itself, which turns changeover duration, rate bands and split days into one idea instead of three special cases.
- Why the physical constraints belong on the tanks and the elastic ones belong on the commercial outcomes — the reverse of the usual first attempt.

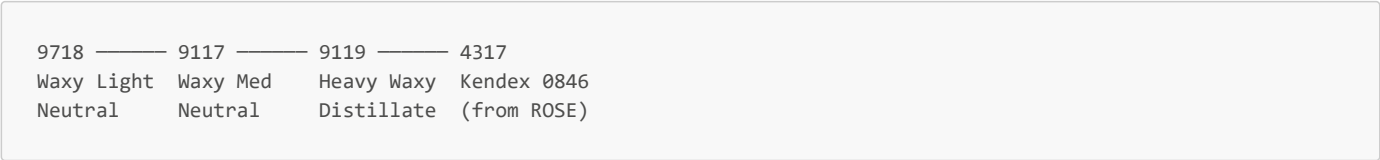
Problem Description

The refinery runs a chain of process units fed by a crude unit whose rate is fixed by refinery economics and is not a decision here. Downstream of it, the MEK unit dewaxes four different waxy stocks. Each produces one dewaxed oil that feeds the next unit in the chain, plus a slack wax that is sold or downgraded:

Charge	Name	Dewaxed oil (yield)	Slack wax (yield)	Goes on to
9718	Waxy Light Neutral	9720 (0.815)	4449 (0.185)	Hydrotreater R2
9117	Waxy Medium Neutral	9302 (0.825)	4451 (0.175)	Extraction
9119	Heavy Waxy Distillate	9303 (0.810)	4454 (0.190)	Extraction
4317	Kendex 0846	9305 (0.840)	4459 (0.160)	Extraction

Three of those charges arrive from the crude unit as fixed supply. The fourth, 4317, is made by the ROSE unit — so the MEK unit both feeds and is fed by the rest of the plant, and cannot be scheduled in isolation.

The unit cannot move freely between charges. Over 366 days of the current plan it made 56 changeovers, and every one moved exactly one rung on a viscosity ladder:



Six of the twelve conceivable transitions are ever used, and they are exactly the adjacent pairs in both directions. A changeover costs an eighth of a day of unit time while the interface is returned to charge; no material is lost.

Tanks are real. Inventory can neither go below zero nor above capacity. What varies is what the refinery *pays* to stay inside those limits: a tank about to overflow is relieved by downgrading to a low-value outlet, and demand that cannot be served is a lost sale. Both are decisions with a price, and the model chooses between them.

The task is to decide, for each of the 42 days, which stock the MEK unit is set up for, how much of it to charge, and where any resulting overflow goes — at least cost, and without ever asking a unit to charge feed that is not in the tank.

Model Formulation

Sets and Indices

$d \in D = \{1, 2, \dots, 42\}$: Days of the detailed horizon, opening 2026-07-23.

$q \in Q$: Set of all model products (45).

$q \in Q^T \subset Q$: Tanked products — those with a physical tank (43).

$q \in Q^F \subset Q$: Untanked flow-through intermediates, made and consumed inside a cascade with no storage anywhere.

$p \in P = \{9718, 9117, 9119, 4317\}$: Charge stocks the MEK unit may run.

$a = (p, p') \in A$: **Allowed changeover arcs**. Only the six adjacent pairs on the ladder exist: $A = \{(9718, 9117), (9117, 9718), (9117, 9119), (9119, 9117), (9119, 4317), (4317, 9119)\}$.

$\ell \in L$: Charge lines on every unit other than MEK, whose assignment stays fixed as in `v0`.

$k \in K = \{6 \text{ oil}, \text{cat}, \text{diesel}, \text{gasoline}\}$: Downgrade outlets.

$R_q \subset K$: Outlets product q may be routed to.

$D^{\text{down}} \subset D$: Days the MEK unit is in turnaround. Two, in this window.

Parameters

$\text{GAL} = 42$: Gallons per barrel.

$\text{rate}_p \in \mathbb{R}^+$: Maximum charge rate (in bbl/day) of stock p — 9718: 4,000; 9117: 4,000; 9119: 2,100; 4317: 2,400.

$\text{loss}_a \in \mathbb{R}^+$: Unit time (in days) consumed by changeover a . 0.125 for every arc on this unit, measured three ways independently.

$L_p \in \mathbb{N}$: Minimum run length (in days) once stock p is started — 9718: 2; 9117: 2; 9119: 1; 4317: 3.

$\text{yld}_{p,q} \in \mathbb{R}^+$: Gallons of product q produced per barrel of stock p charged, i.e. the yield fraction times GAL.

$\text{yld}_{\ell,q}^L \in \mathbb{R}^+$: The same, for the fixed lines.

$\text{cap}_q \in \mathbb{R}^+$: Tank capacity (in gallons) of product q .

$I_q^0 \in \mathbb{R}^+$: Opening inventory (in gallons).

$\text{dem}_{q,d} \in \mathbb{R}^+$: Demand (in gallons) — firm orders plus netted forecast plus blend-component pull.

$\text{sup}_{q,d} \in \mathbb{R}^+$: Fixed supply (in gallons) arriving from the crude unit, which is an input rather than a decision.

$\text{plan}_{\ell,d} \in \mathbb{R}^+$: Barrels the planner scheduled on fixed line ℓ .

$\phi \in [0, 1]$: Charge floor — the fraction of the planner's own charge each fixed line must still run.

$\sigma_p^0 \in \{0, 1\}$: Day-zero setup state, 1 for the stock the unit is actually holding when the horizon opens.

$m_q \in \mathbb{R}^+$: Lost-sale margin (in USD/gal).

$c_k^{\text{dg}} \in \mathbb{R}^+$: Margin given up (in USD/gal) by routing to outlet k .

$c^{\text{term}} \in \mathbb{R}^+$: Cost (in USD/gal) of ending the window below opening inventory.

$c^{\text{sw}} \in \mathbb{R}^+$: Cash cost (in USD) of a changeover — labour, quality giveaway, and anything else the lost time does not already capture.

Decision Variables

$x_{p,d} \in \{0, 1\}$: 1 if the MEK unit is **set up for** stock p at the end of day d . Persists through idle days.

$z_{a,d} \in \{0, 1\}$: 1 if changeover a is made during day d . Defined only for $a \in A$.

$y_{p,d} \in \{0, 1\}$: 1 if stock p receives any of day d 's time.

$\tau_{p,d} \in [0, 1]$: Fraction of day d spent charging stock p .

$g_{p,d} \in \mathbb{R}^+$: Barrels of stock p charged to MEK on day d .

$g_{\ell,d}^L \in \mathbb{R}^+$: Barrels charged on fixed line ℓ .

$w_{q,k,d} \in \mathbb{R}^+$: Gallons of q routed to outlet k .

$u_{q,d} \in \mathbb{R}^+$: Gallons of demand for q left unserved.

$v_{q,d} \in \mathbb{R}^+$: Gallons sold down from a sink beyond its own forecast.

$I_{q,d} \in [0, \text{cap}_q]$: Ending inventory (in gallons).

$e_q \in \mathbb{R}^+$: Gallons by which the window ends below opening inventory.

Assumptions

1. **Crude rate is an input, not a decision.** Its margin dwarfs anything downstream, so an optimizer free to cut crude would relieve every inventory problem by making less oil.
2. **At most two changeovers a day**, which is exact on 364 of 366 observed days.
3. **The interface returns to charge**, so a changeover costs time and not material.
4. **Charge is continuous within the rate band.** Only 12.8% of charged unit-days in the plan sit exactly on the published run rate, so the level is a decision.
5. **Minimum charge rates are zero.** A minimum is a restriction; starting without one guarantees the model has a feasible answer, and adding real numbers later can only tighten the plan.
6. **The MEK charges workbook product 9116, whose inventory lives in the block keyed 9718** — the two share a tank. Resolving to the wrong key would let the model consume stock it never decremented.

Objective Function

Cost of relief. Minimise what the refinery pays to keep every tank inside its limits over the horizon (in USD).

$$\text{Min } Z = \sum_{d \in D} \sum_{q \in Q^T} m_q u_{q,d} + \sum_{d \in D} \sum_{q \in Q^T} \sum_{k \in R_q} c_k^{\text{dg}} w_{q,k,d} + \sum_{d \in D} \sum_q c_{\text{flat}}^{\text{dg}} v_{q,d} + \sum_{q \in Q^T} c^{\text{term}} e_q + \sum_{d \in D} \sum_{a \in A} c^{\text{sw}} z_{a,d} \quad (0)$$

Every term is denominated in dollars, so no penalty weights need tuning and no artificial priority tiers are required. The changeover term carries only the cash a switch costs beyond the time it consumes; the time itself is priced by the production it displaces, through constraint (7).

Note. Once per-product netbacks exist, this objective inverts to a margin maximisation and the first four terms become revenue forgone rather than penalties assessed. See [MARGIN-OBJECTIVE.md](#). The constraint set below does not change.

Constraints

Setup carryover. The stock the unit is set up for at the end of a day is what it was set up for yesterday, adjusted by any changeover made today. This is what makes the setup persist through an idle day.

$$x_{p,d} = x_{p,d-1} + \sum_{(p',p) \in A} z_{(p',p),d} - \sum_{(p,p') \in A} z_{(p,p'),d} \quad \forall (p,d) \in P \times D \quad (1)$$

Exclusive setup. The unit is set up for exactly one stock at all times, idle or not.

$$\sum_{p \in P} x_{p,d} = 1 \quad \forall d \in D \quad (2)$$

Day-zero state. The horizon opens with the unit holding what it is actually holding, supplied by the live feed rather than by the schedule.

$$x_{p,0} = \sigma_p^0 \quad \forall p \in P \quad (3)$$

Changeover limit. At most two changeovers may be made in a day.

$$\sum_{a \in A} z_{a,d} \leq 2 \quad \forall d \in D \quad (4)$$

Time follows setup. A stock may only receive time on a day it is set up for, either at the start of the day or at the end of it — which is what allows a changeover day to carry both the outgoing and the incoming charge.

$$y_{p,d} \leq x_{p,d-1} + x_{p,d} \quad \forall (p,d) \in P \times D \quad (5)$$

$$\tau_{p,d} \leq y_{p,d} \quad \forall (p,d) \in P \times D \quad (6)$$

Day-time budget. The day's time is shared between charging and changing over. This is where a changeover is paid for.

$$\sum_{p \in P} \tau_{p,d} + \sum_{a \in A} \text{loss}_a z_{a,d} \leq 1 \quad \forall d \in D \quad (7)$$

Must run. Outside a declared turnaround the unit runs. Without this the optimizer would relieve any inventory problem by simply not running, which is never what the plant does.

$$\sum_{p \in P} y_{p,d} \geq 1 \quad \forall d \in D \setminus D^{\text{down}} \quad (8)$$

$$y_{p,d} = 0 \quad \forall (p,d) \in P \times D^{\text{down}} \quad (9)$$

Charge rate. Barrels charged are the rate times the time spent, which is linear because time is the variable.

$$g_{p,d} \leq \text{rate}_p \tau_{p,d} \quad \forall (p,d) \in P \times D \quad (10)$$

Minimum run length. A stock that is started must stay on the unit for its minimum campaign.

$$\sum_{t=d}^{\min(d+L_p-1, |D|)} x_{p,t} \geq L_p \sum_{(p',p) \in A} z_{(p',p),d} \quad \forall (p,d) \in P \times D \quad (11)$$

Fixed-line throughput floor. Every other unit still runs at least the given fraction of what the planner scheduled. The objective counts only costs, so without a floor the cheapest schedule elsewhere is to stop making oil.

$$\phi \cdot \text{plan}_{\ell,d} \leq g_{\ell,d}^L \leq \max(\text{rate}_{\ell}, \text{plan}_{\ell,d}) \quad \forall (\ell,d) \in L \times D \quad (12)$$

Inventory balance. For every tanked product on every day, in gallons.

$$I_{q,d} = I_{q,d-1} + \sum_{p \in P} \text{yld}_{p,q} g_{p,d} + \sum_{\ell \in L} \text{yld}_{\ell,q}^L g_{\ell,d}^L + \text{sup}_{q,d} - \text{GAL} \sum_{p \in P: p=q} g_{p,d} - \text{GAL} \sum_{\ell \in L: \ell \text{ charges } q} g_{\ell,d}^L - \text{dem}_{q,d} + u_{q,d} - \sum_{k \in R_q} w_{q,k,d} \quad (13)$$

$$\forall (q,d) \in Q^T \times D$$

Consumption must sum **every** line that charges q . The workbook subtracts only the first, which is a defect the optimizer must not inherit or it will plan on feed that does not exist.

Tank limits. Hard on both sides, as variable bounds.

$$0 \leq I_{q,d} \leq \text{cap}_q \quad \forall (q,d) \in Q^T \times D \quad (14)$$

Lost sales are bounded by demand. You cannot fail to serve more than was asked for.

$$u_{q,d} \leq \text{dem}_{q,d} \quad \forall (q,d) \in Q^T \times D \quad (15)$$

Flow-through identity. An intermediate with no tank has no buffer, so everything made of it is charged the same day. This replaces a rate ceiling on the downstream unit rather than supplementing one.

$$\sum_{\ell \in L} \text{yld}_{\ell,q}^L g_{\ell,d}^L = \text{GAL} \sum_{\ell \in L: \ell \text{ charges } q} g_{\ell,d}^L \quad \forall (q,d) \in Q^F \times D \quad (16)$$

Terminal inventory. The window hands the next one a plant in the state it found it. Elastic, because demand genuinely can exceed supply and that is a situation the planner needs an answer for rather than a solver error.

$$I_{q,|D|} + e_q \geq I_q^0 \quad \forall q \in Q^T \quad (17)$$

Size

	Count
Setup binaries x	168
Arc binaries z	252
Time-allocation binaries y	168
Total binaries	588
Continuous variables	~2,300

Six arcs rather than twelve is the point of the formulation. Forbidden transitions are not constrained away — **they do not exist**, so the solver cannot reach for them and the LP relaxation is not weakened by excluding them.

Python Implementation

The model is built with PuLP and solved with CBC. It lives beside `v0`:

```
src/invplanner/optimizer/
v0.py      LP: charge levels and downgrade routing, assignments fixed
v1.py      + setup binaries, changeover arcs and the time budget on MEK
verify.py  replay any result through the simulator and check it holds up
```

Every parameter above is read from the model specification rather than hardcoded:

```
spec = modelprep.build(ref, scn, sim, horizon=dates, downtime=downtime)
unit = spec.units["MEK"]

unit["products"]          # ['9117', '9119', '9718', '4317']
unit["allowed_arcs"]      # the six ladder arcs
unit["min_run_days"]      # {'9117': 2, '9119': 1, '9718': 2, '4317': 3}
unit["max_rate_by_line"]  # bbl/day, per line
unit["changeover_loss_days"] # 0.125 on every arc
```

The arc set comes from `model_config.UNIT_TRANSITIONS`, which was measured from the plan and is meant to be confirmed with operations rather than trusted.

Model Deployment

The model runs behind the planning application. A planner sets the economic inputs and the planned downtime, presses **Run**, and the result lands as a new scenario they can open, edit and compare against their own.

```
python scripts/run_api.py      # then Optimizer → Run on the baseline scenario
```

Each run is replayed through the simulator — an independent implementation of the same physics, validated cell-by-cell against the workbook — and is marked `unverified` if the two disagree. **The optimizer's own numbers are computed from its own variables, so they are exactly what not to trust.**

Analysis

A run is worth acting on only if all of the following hold.

Schedule shape

The formulation imposes adjacency and a minimum campaign; it does not impose the sweep up and down the ladder that the plant actually runs. **That sweep should emerge on its own.** If it does not, the transition data is wrong rather than the solver.

Measure	Plan	Model should be near
Campaigns over 42 days	~7	comparable
Median campaign length	3 days	2–5 days
Switches per month	4.6	operable, per planners
Days lost to changeovers	~0.8	falls out of (7)

Cost

The result must beat the plan on the measures that motivated the work, scored the same way on both sides:

Plan, 42 days	
Lost sales	3,207,855 gal
Downgrade	3,398,560 gal
Feed shortfall	870,449 gal

Executability

Feed shortfall must be zero. The optimizer holds inventory at or above zero by construction, so any shortfall on replay means the two models disagree about what the schedule consumes — which is a defect, not a cost.

Solve time

`v0` solves in under a quarter of a second. 588 binaries with a warm start from the planner's own schedule should stay well inside a planner's patience; if it does not, the horizon is the lever, not the formulation.

References

1. Williams, H.P. *Model Building in Mathematical Programming*, 5th edition.
2. Gurobi modeling examples: *Factory Planning I & II*, *Food Manufacture I & II* — the source of this document's structure.
3. `MIP-FORMULATION.md` — what the data says the formulation has to be, and the measurements behind every parameter above.
4. `PROCESS-FLOW.md` — the plant as modelled, confirmed with operations.
5. `MARGIN-OBJECTIVE.md` — pricing the objective in real margin.
6. `data/reports/switching-analysis.md` — the stress tests that validated the time-allocation model, the setup-through-idle requirement, and the two- changeover limit.